$R_{XY} = E[XY]$

X	Uniform dist $(-1, 1)$	$E[X] = 0$	$E[X^2] = 1/3$
N	Normal dist. $(0, 1)$	$E[N] = 0$	$E[N^2] = 1$

$E[XN] = E[X] \cdot E[N] = 0 \cdot 0 = 0$

→ $E[x^2]$ for uniform distribution:

$$\int_{-1}^{1} x^2 \frac{1}{2} \cdot dx = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$$

→ $Y = X + N$

$E[XY] = E[X(X+N)]$

$= E[X^2] + E[XN]$

$= \frac{1}{3} + 0$

$\boxed{E[XY] = \frac{1}{3}}$

→ $Z = X^2 + N$

$E[XZ] = E[X^3] + E[XN]$

↳ odd f^n so $\int_{-1}^{1}$ of odd f^n is 0.

$= 0 + 0$

$= 0$

$\boxed{E[XZ] = 0}$